AMITY INSTITUTE OF INFORMATION TECHNOLOGY

WEEKLY PROGRESS REPORT
FOR THE WEEK COMMENCING (08/06/2026-14/06/2026)

WPR No: 05
Program: MCA

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ENROLLMENT NO.- A010145025215

SEMESTER: 3rd
FACULTY GUIDE NAME: Dr. Garima Bohra

PROJECT TITLE: RADMAP – GPS Enabled Radiation Mapping Prototype using Raspberry Pi

TARGETS FOR THE WEEK:

1. Researched radiation detection systems used in nuclear laboratory environments.
2. **LiDAR Sensor Study and Integration**
   Worked on understanding the LiDAR sensor architecture, communication protocol, and planned its integration for distance measurement and scanning operations.
3. **GPS Module (GlobalSat) Testing**
   Targeted successful configuration and testing of the GlobalSat GPS module for retrieving accurate location data through terminal execution.
4. **Gamma Radiation Detector Setup and Software Testing**
   Targeted installation, software configuration, and successful testing of the gamma radiation detector on terminal for radiation data acquisition.
5. **SG90 Servo Motor Connection Establishment**
   Planned hardware connection and testing of the SG90 servo motor for movement control and directional scanning operations.

ACHIEVEMENTS OF THE WEEK:

1. Completed literature review on radiation monitoring and hazardous environment systems.
2. **GPS Module Successfully Running on Terminal**
   Configured and successfully executed the GlobalSat GPS module on terminal, verifying communication and proper location data output.
3. **Gamma Radiation Detector Successfully Tested**
   Installed supporting software, established hardware connection, and successfully ran the gamma radiation detector on terminal with proper response verification.
4. **Raspberry Pi Environment Prepared Successfully**
   Completed Raspberry Pi setup and confirmed readiness for integration of sensors and hardware modules through terminal operations.

FUTURE WORK PLANS:

1. **Hardware Integration with Raspberry Pi**
   Integrate all core hardware components including Raspberry Pi, LiDAR Sensor, SG90 Servo Motor, and ESP8266 camera module into a unified system architecture.
2. **Final Hardware Integration**
   Integrate LiDAR sensor, GlobalSat GPS module, Raspberry Pi, Servo Motor, and Gamma Radiation Detector into a unified system architecture.
3. **Communication Synchronization Between Modules**
   Establish proper communication channels between all hardware components for synchronized and stable operation.
4. **Real-Time Data Collection Implementation**
   Begin implementation for collecting live sensor data simultaneously from GPS, radiation detector, and LiDAR system.
5. **Testing and Troubleshooting of Integrated System**
   Perform debugging, troubleshoot hardware/software integration issues, and optimize component communication.
6. **End-to-End System Deployment and Validation**
   Conduct final testing of the complete integrated system to validate overall functionality and prepare for deployment stage.